CHILDREN'S DISCOVERY

THE SCIENCE OF PLAY

GREEN LIGHTING OUR WORLD

GLOW

GLOW House



QUICK START GUIDE

RECOMMENDED AGES 7+

READY IN 5 MIN



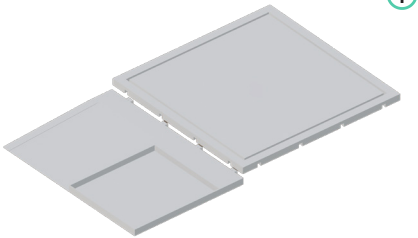
Hi, I'm Lumi! Follow me from box to bright house.

Adult supervision is required at all times. Small parts - not suitable for children under 3 years old. Always read the instructions.

CHECK EVERY PART

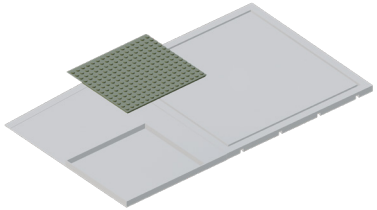
WHAT'S IN THE BOX

1



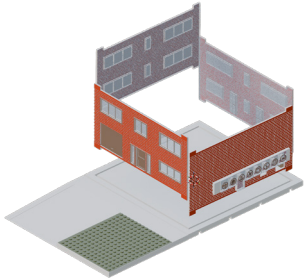
2x BASE PLATES

2



1x LAWN PLATE

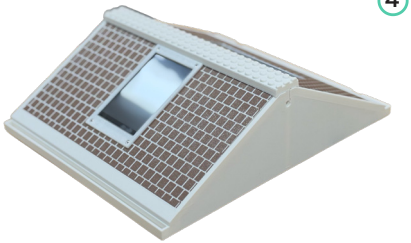
3



4x WALLS

Utility wall comes ready-built.

4



1x ROOF

The panel and screen already fitted.

5



1x POWER USB-C CABLE

6



1x EV USB-C CABLE

ADD-ON **Got the EV car and RC controller? They have their own guide.**

GET READY

BEFORE YOU START

1

Clear a table

Make space for every part.

2

Grown-up helps

Ask an adult to help with the cables.

3

Find a power source

Any standard USB-C charger. Even a smartphone's USB-C port.

4

Keep the packaging

Everything packs back in.

YOUR JOURNEY

BUILD

Steps 1–7 · page 4

POWER

Steps 8–11 · page 9

PLAY

Home screen and 10 challenges · pages 12–16

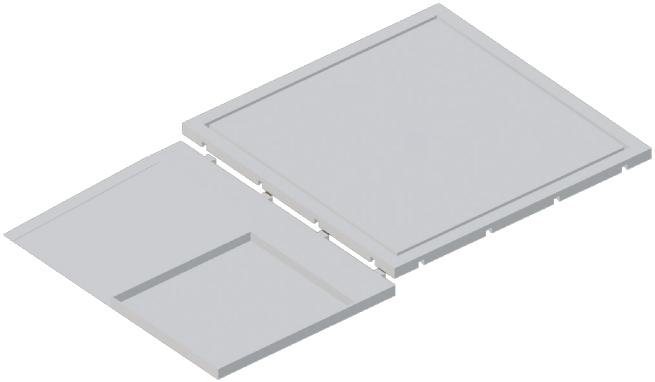


READY? LET'S BUILD!

1 JOIN THE BASE PLATES

YOU NEED
2×
Base plates

Press the two plates together.

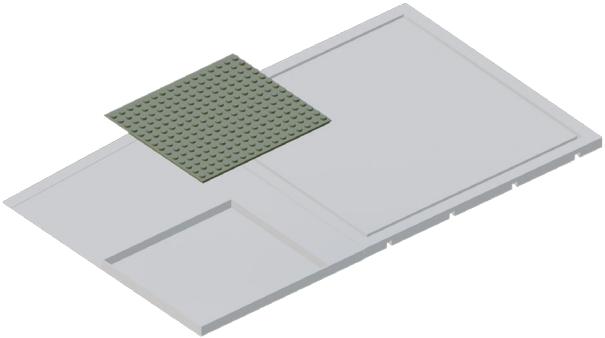


Line up the edges first.

2 PRESS IN THE LAWN

YOU NEED
1×
Lawn plate

Press down until it clicks.

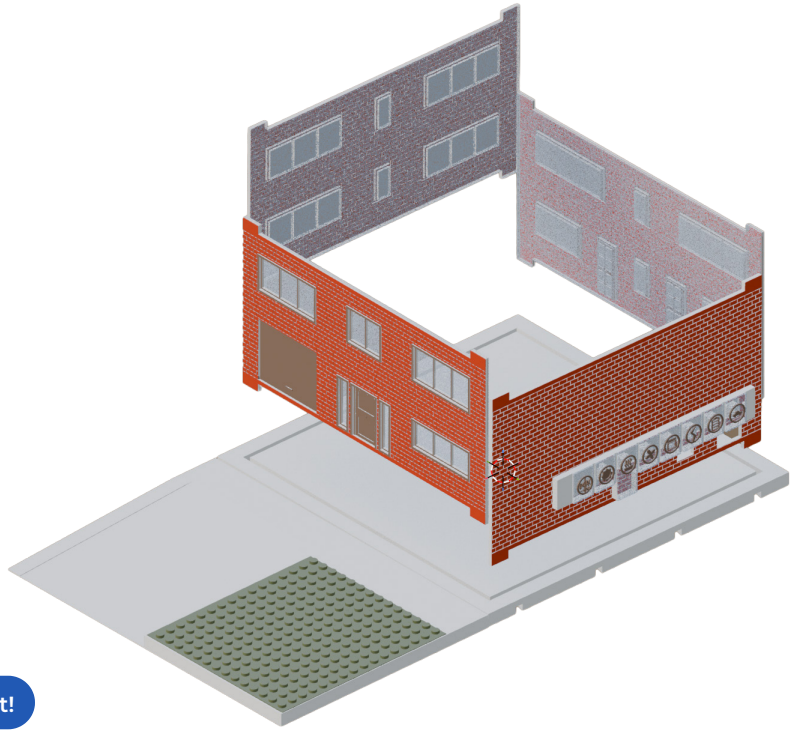


Listen for the click!

3 Snap on the four walls

Any order. Press each one down.

YOU NEED
4x
Walls



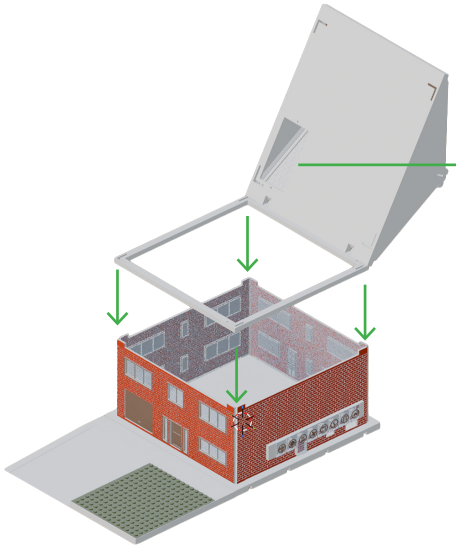


Brick side faces out!

4 PLACE THE ROOF

Lower it onto the walls.

YOU NEED
1x
Roof

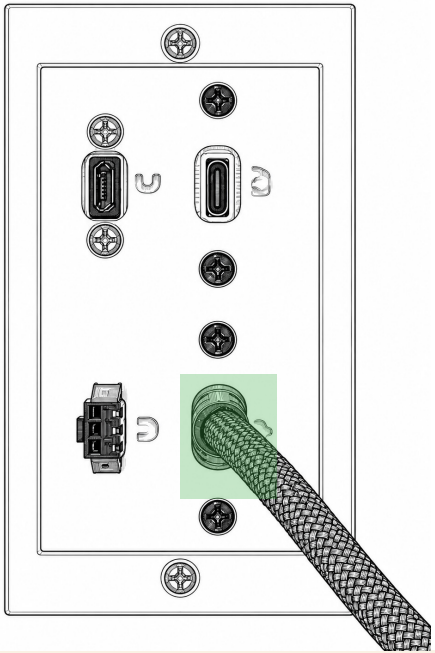


Press down at both ends.

5 PLUG IN THE POWER CABLE

Plug the power USB-C into the roof panel.

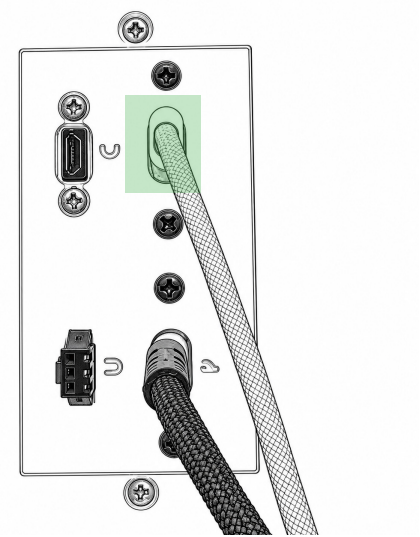
YOU NEED
1x
Power USB-C cable



6 PLUG IN THE EV CABLE

YOU NEED
1x
EV USB-C cable

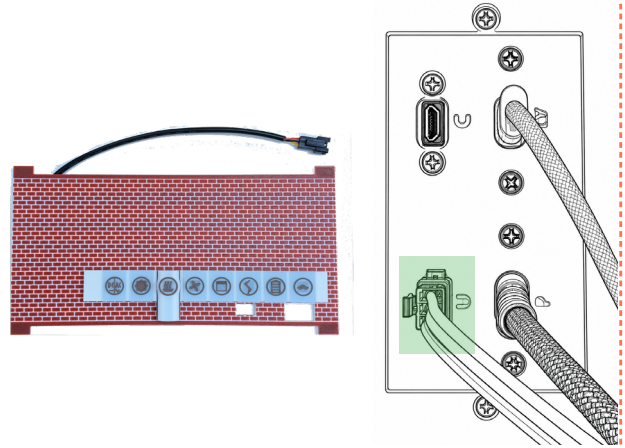
Plug the EV USB-C cable into the roof panel.



Two cables in. Nice!

7 PLUG IN THE WALL LIGHTS

into the roof panel.



The cable comes out of the utility wall.

This lights up the wall strip.

CHECKPOINT

YOU BUILT IT!

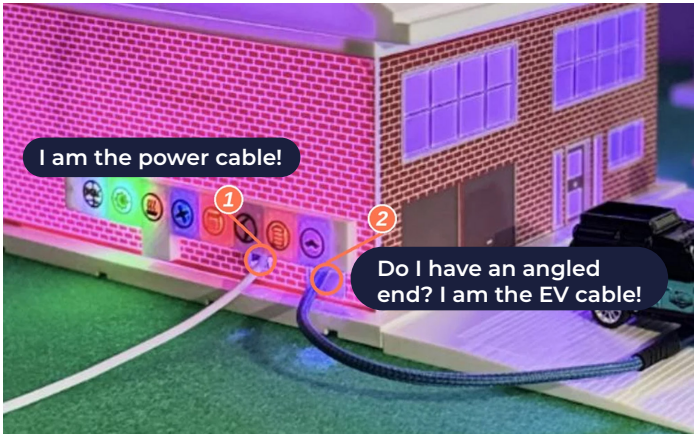
Check it over before you power up.

- ✓ Three cables plugged in
- ✓ Walls snapped in
- ✓ Roof sits flat
- ✓ No cable pinched



8 CONNECT THE POWER

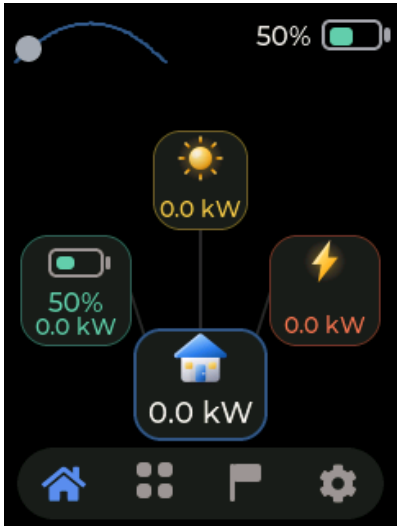
Plug the other end into any USB-C power source.



Plug the other end of the power cable to power source and watch the magic happen!

9 WAIT FOR THE SCREEN

The home screen appears.



Hello, house!

10 COVER THE SOLAR PANEL

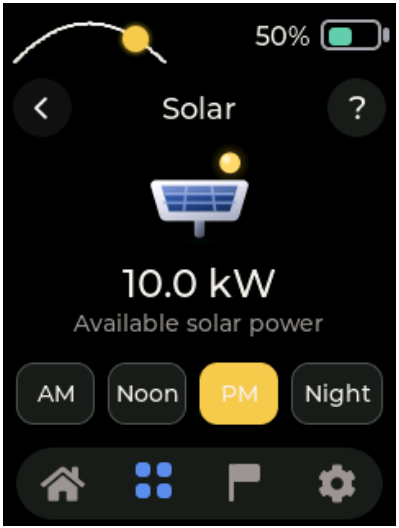
Cover it with your hand. Watch the kW drop.



The panel is real!

11 DIM THE ROOM

Less light, fewer kW.



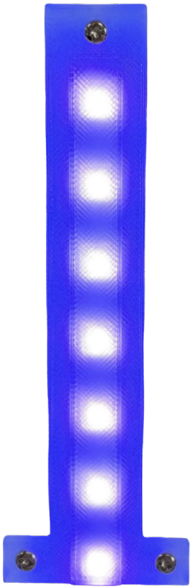
Less light, less power.

READING THE LIGHTS



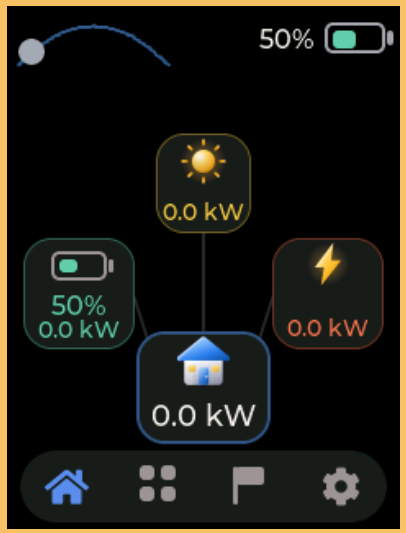
WALL STRIP

Lights up when you switch things on.



CEILING LIGHT

Inside, like a real ceiling light.



Meet the display

Four buttons along the bottom, left to right.

1. **HOME**
   See the sun, battery, grid and house.

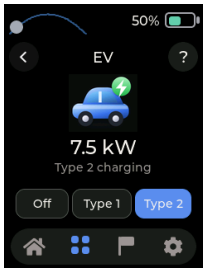
2. **CONTROL**
   Pick something to switch or slide.

3. **CHALLENGES (THE FLAG)**
   Start any of the 10. Complete them all.

4. **SETTINGS**
   Recalibrate the touch screen. Change the simulation speed. Reset the challenges and settings.

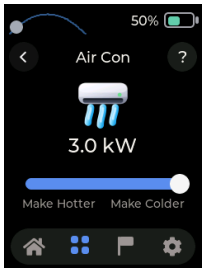
8 THINGS TO CONTROL

EV



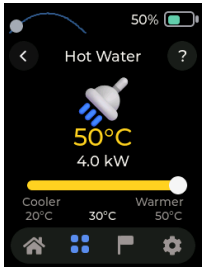
Off, Type 1 or Type 2.

AIR CON



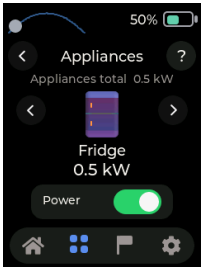
Slide hotter or colder.

HOT WATER



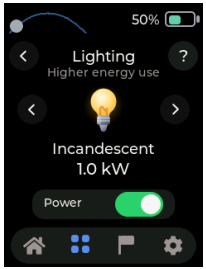
Slide from 20 to 50°C.

APPLIANCES



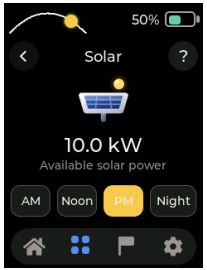
Switch the fridge, oven, TV and more.

LIGHTING



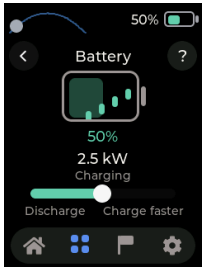
Pick a light. Switch it on.

SOLAR



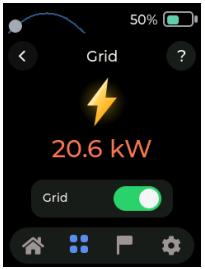
Pick AM, Noon, PM or Night.

BATTERY



Charge it, or let it discharge.

GRID



Switch the grid on or off.

REAL-FAMILY SCENARIOS

10 ENERGY CHALLENGES

1

Tap the flag

Pick a challenge. Read what it is.

2

Press start

The house enters simulation mode.

3

Spot the green dot

Top middle of the screen. You're in a challenge.

4

Fill the bar

A progress bar tracks you. It finishes by itself.

LEVELS STARTER → MODERATE → ADVANCED → EXPERT

1 STARTER

MORNING POWER RUSH

7 AM. Solar rising. Battery at 60%.

YOUR GOAL
Keep grid draw below 2 kW as the family wakes up.

Energy sources · Cause & effect ○ DONE

2 STARTER

SOLAR POWER CHAMPION

12 PM. Clear sky.

YOUR GOAL
Zero grid draw for five minutes in a row, on solar alone.

Renewable energy · Data observation ○ DONE

3 MODERATE

NIGHT SURVIVAL

8 PM. Solar offline. Battery at 80%.

YOUR GOAL
Power the whole evening on the battery. Keep the grid at zero.

Energy storage · Strategic planning ○ DONE

4 MODERATE

ELECTRIC CAR RACE

The EV is at 20%. Three charging plans compete.

YOUR GOAL
Charge to 80% at the lowest possible cost.

Electrification · Financial maths ○ DONE

5 ADVANCED

HEATWAVE SURVIVAL

It's 38°C. The AC must run.

YOUR GOAL
Keep the grid below 1.5 kW as the battery runs down.

Demand management · Systems thinking ○ DONE

How to play a card

1. Tap the flag. Pick the number.
2. Read it. Press start.
3. Reach the goal.
4. Bar full? You've completed the challenge! DONE.

Frustration is a feature. Try again!

6 MODERATE

BIRTHDAY PARTY POWER

3 PM. 90% solar. Party time!

YOUR GOAL
Run the party loads. Push the extra solar into the battery.

Load management · Optimisation ○ DONE

7 MODERATE

BLACKOUT EMERGENCY

The grid goes down. Battery at 55%.

YOUR GOAL
Choose what stays on. Survive on solar and battery alone.

Grid resilience · Decisions under pressure ○ DONE

8 ADVANCED

ZERO EMISSIONS DAY

A 24-hour simulation. Every decision counts.

YOUR GOAL
Zero grid draw from dawn to dusk.

Carbon emissions · Extended planning ○ DONE

9 ADVANCED

MONEY-SAVING MARATHON

A week-long simulation.

YOUR GOAL
Cut the household bill by at least 50% with smart scheduling.

Financial literacy · Data analysis ○ DONE

10 EXPERT

ULTIMATE ENERGY MASTER

A full day of weather, surprise loads and a grid outage.

YOUR GOAL
Get through the day. You'll need every skill.

Integrated systems · Creative problem-solving ○ DONE

LEVEL KEY

STARTER	1 · 2
MODERATE	3 · 4 · 6 · 7
ADVANCED	5 · 8 · 9
EXPERT	10

Tick all ten? Make the planet smile.

DIGITAL COMPANION

STEP INSIDE ONLINE

Explore the rooms.

Tap the appliances.

Learn how it all works.

childrensdiscovery.org.au/glow

Need help?

SCREEN BLANK?
Check both ends of the power cable.

WALL STRIP DARK?
Check the wall LED cable is plugged into the panel.

TOUCH FEELS OFF?
Recalibrate the screen in Settings.

Still stuck? Get in touch

info@childrensdiscovery.org.au

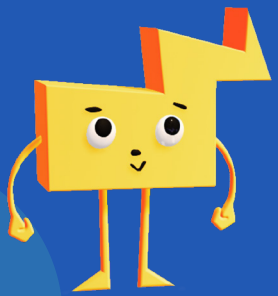
CHILDREN'S DISCOVERY

CHILDREN'S DISCOVERY

THE SCIENCE OF PLAY

GLOW House





Thank you for purchasing the GLOW House!